

Forward jets, missing p_T and dijets from every vertex in high pileup: a jet-vertex association by global missing- p_T minimisation, tested against the truth

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<https://github.com/miquork/genseed>

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Abstract

PUPPI cannot tell vertices apart beyond the tracker: every particle-flow candidate with $|\eta| > 2.5$ gets the same weight at every vertex. On a NeutrinoGun sample with $\langle N_{PU} \rangle = 45$ and the generated particles of every interaction, a per-vertex reconstruction therefore carries the same forward jets at every vertex: **50 times** the generated jets of the vertex's own interaction at $2.5 < |\eta| < 3.0$ above 10 GeV, made in 71% of the cases of several interactions at once, and a missing- p_T of 15 GeV rms per component against 3 GeV of true imbalance. We implemented the association of the forward energy to vertices by a global minimisation of the summed missing- p_T χ^2 over all vertices (JVA), with an exact, deterministic optimiser, and tested it against the generator truth of every interaction. **It does not work at these momenta:** the forward energy that belongs to one interaction carries 1.6 GeV per vertex against 7 GeV of recoil noise, every recoil definition (particle missing- p_T , or the missing- H_T of jets above 3–15 GeV) correlates with it at $r = 0.13$ – 0.16 , and the best configuration places 1.4% more single-interaction clusters right than wrong. Two results do work. **Extending PUPPI's charged-vertex association and its α from $|\eta| < 2.5$ to $|\eta| < 3.0$,** using the charged candidates that exist there, cuts the 2.5–3.0 spike by a factor 25–80 at every vertex (reco/gen above 10 GeV $49.6 \rightarrow 2.0$) and the missing- p_T at the primary vertex from 17.0 to 15.1 GeV. And **dijets selected at every vertex** give 130 times more low- p_T balanced dijets per crossing than the primary vertex alone; their bisector width reproduces the true jet resolution in the barrel from $p_T^{\text{avg}} = 5$ GeV. The L3 residual needs Z+jet events, which this sample does not have.

1 The problem

genseed reconstructs every usable vertex v (34 per crossing, 31 of them owned by an interaction) with PUPPI weights $w_i(v)$. Inside $|\eta| < 2.5$ a charged candidate belongs to a vertex and a neutral one is weighted by the charged activity of that vertex around it. Beyond, PUPPI's weight w_i^{fwd} does not depend on v . Measured on the sample (300 crossings):

- every vertex carries 3.6 forward jets above 5 GeV at $2.5 < |\eta| < 3.0$ and 1.7 at $3 < |\eta| < 5$, 98–100% of their p_T from vertex-blind candidates; their cone holds 4.2 tracker tracks, 4.1 of them associated to other vertices;
- from the genseed links, the dominant interaction of such a jet carries less than 40% of its linked generated p_T in 71% of the jets and more than 60% in 7%; three interactions contribute more than 10% each. The spike is pileup made of several interactions, not misplaced jets;
- the particle missing- p_T per vertex has 25 GeV rms per component at the primary vertex, against 3.2 GeV of true imbalance of an interaction inside $|\eta| < 5$;
- the missing- p_T of the vertex-resolved part ($|\eta| < 2.5$) scatters around the owner's true central recoil by $\sigma \simeq \sqrt{1 + S_v}$ GeV per component, S_v the scalar sum of its weighted p_T (5.8 GeV at $S_v = 30$), while the true forward p_T of an interaction is 4.8 GeV rms.

2 The jet-vertex association

Objects. The vertex-blind candidates ($|\eta| > \eta_{\text{fwd}} = 2.5$, $w^{\text{fwd}} > 0$) are clustered once per crossing (anti- k_T , $R = 0.4$, no threshold) into forward clusters F_k with transverse momentum $\vec{p}_k$: every such candidate is in exactly one cluster (24 per crossing). For each vertex, $\vec{M}_v = -\sum w_i(v) \vec{p}_{T,i}$ over its vertex-resolved candidates.

Objective. An assignment $h_k \in \{\text{vertices}\} \cup \{\emptyset\}$ minimises

$$X(h) = \sum_v \frac{|\vec{M}_v - \sum_{k: h_k=v} \vec{p}_k|^2}{\sigma_v^2} + \sum_{k: h_k \neq \emptyset} \tau_k, \quad \sigma_v^2 = \sigma_0^2 + \sigma_K^2 S_v, \quad (1)$$

so that a cluster is placed only where it improves the χ^2 by more than its price τ_k , and a cluster that balances no vertex — a combination of interactions — goes to nobody. Candidates: the vertices whose tracks lie in the cluster (in the cone of the tracker tracks, `candMode 0`; its own charged constituents with $2.5 < |\eta| < 3$, whose vertex reference PUPPI ignores, `candMode 1`; either, `candMode 2`) at price τ_{trk} , or any vertex at price τ_{all} for a cluster without candidates. Recoil: the particle missing- p_T above (`recoilMode 0`) or the missing- H_T of the vertex-resolved jets above T (`recoilMode 1`, $\sigma_v^2 = \sigma_0^2 + \sigma_J^2 \sum p_T^2 + \sigma_K^2 (S_v - H_T)$).

Optimiser (`javassoc.h`, ROOT-free). Clusters interact only through shared vertices, so the problem splits exactly into connected components (union-find over shared candidates); a component is solved exhaustively over the product of its options up to 2×10^5 combinations, otherwise by coordinate descent from the best single moves with pairwise re-optimisation. On random instances the exhaustive path equals brute force in every case and the descent reaches the optimum in 98.6%; on the data, 30 perturbed restarts per crossing never found a lower objective. The result is deterministic (ties to $\emptyset$, then to the lowest vertex).

Methods compared (each gives an event for every vertex, which is clustered and whose missing- p_T is taken): `dup` — every vertex keeps all forward candidates (per-vertex PUPPI as it is, and CMS PUPPI at the primary vertex); `1v` — the forward candidates go to the primary vertex only (the leading-vertex approach); `trk` — a cluster goes to the candidate with the most track evidence; `jva`; `none` — the forward energy goes to nobody; and two oracles from the links, `ojet` (a cluster to the vertex of its dominant interaction when that holds more than half of its linked p_T , else nobody) and `opart` (every candidate to its own dominant interaction’s vertex). Knobs, histograms and tuple: `JVA.h`.

3 Results

All numbers: 25 000 crossings (856 527 vertices), `rootfiles/JVA_v2.root`; the full tables and figures are in the appendix.

3.1 Missing p_T

Per component, rms over all vertices / at the primary vertex, in GeV: `dup` 14.90 / 17.03, `1v` 7.18 / 17.03, `jva` 6.89 / 13.22, `none` 6.96 / 13.55, `ojet` 6.92 / 12.96, `opart` 6.88 / 12.76 (Table ??). **Removing the forward duplicates halves the missing- p_T width over all vertices**, and at the primary vertex dropping the forward energy (`none`) is as good as any association. The width is **not a figure of merit** for a method that minimises it: lowering the prices pushes `jva` below the oracles by fitting noise, and even the residual against the owner’s true imbalance is circular, because the truth (3.2 GeV) is small against the noise (7 GeV).

3.2 The association against the truth

With the default prices ($\tau_{\text{trk}} = 4$, $\tau_{\text{all}} = 12$) `jva` places 0.2% of the forward clusters above 5 GeV right and drops nearly everything; its correct fraction (right plus combinations dropped) is 0.741 against 0.762 for `none`. The offline re-optimisation from the tuple, `scanJVA.C`, over the recoil (the particle missing p_T , or the missing H_T of the jets above 3, 5, 7, 10 or 15 GeV), the candidates, the prices and the resolution scale, finds at best a net $+415 \pm 42$ of 29 899 single-interaction clusters right (1.4%; the missing H_T of jets above 15 GeV, in-cluster charged candidates), while keeping 9.7% of the combinations and assigning single-interaction clusters and combinations at the same rate: the balance does not know which vertex. The reason is quantitative:

recoil R_v	$\vec{M}_v$	$H_T^{\text{miss}}, T = 3$	5	7	10	15 GeV
correlation with the owner's true forward p_T	0.457	0.396	0.345	0.298	0.239	0.176
correlation with what the clusters can supply, F_v	0.138	0.133	0.133	0.136	0.144	0.159

The missing- H_T of low-threshold jets is worse than the particle missing- p_T at every threshold: the recoil of forward minimum-bias activity is soft (even the generated jets above 10 GeV carry only $r = 0.35$ of it). And the forward p_T that PUPPI leaves and that belongs to one interaction, F_v , is only 1.6 GeV per component per vertex (its forward floor, $wp_T > 1.85 + 0.070 N_{\text{PV}} = 3.9$ GeV, removes most of it): nothing that measures a recoil of 7 GeV noise can find it. Tracks alone do better: the vertex holding most of a cluster's own charged p_T is the right one for 31% of the single-interaction clusters above 5 GeV (in-cone tracker tracks: 7%), and assigning by it (else nobody) gives a net $+5324 \pm 110$ (17.8%), but keeps 77% of the combinations.

3.3 The forward spike, and PUPPI with tracks to $|\eta| = 3$

Reconstructed over generated jets per owned vertex at $2.5 < |\eta| < 3.0$ above 10 GeV: `dup` 49.6, `1v` 1.53 (all of it at the primary vertex), `jva` 0.10, `ojet` 0.24, `opart` 0.15, `none` 0: the spike disappears only by dropping, and even the oracles recover a quarter of the generated jets. The handle is the charged candidates at $2.5 < |\eta| < 3$: giving PUPPI's vertex association and its α the same reach (`pupEtaVtxAssoc = pupEtaTracker = etaFwd = 3`, `cmpPuppiFwd.C`):

PUPPI	reco/gen >5	>10	>20 GeV	own fraction	missing- p_T rms, all / PV
default (vertex-blind beyond 2.5), <code>dup</code>	22.5	49.6	64.7	0.03	14.90 / 17.03
tracks and α to $ \eta < 3$, <code>dup</code>	1.07	1.99	0.80	0.18	11.56 / 15.06
same, central neutral floor in 2.5–3	1.58	2.77	1.20	0.18	11.73 / 15.24

The residual excess sits at $2.7 < |\eta| < 2.9$ (2.6 times the generated jets above 5 GeV), and the region beyond $|\eta| = 3$, where no track exists, keeps its own spike (4–8 times at $3.0 < |\eta| < 3.3$ with `dup`). The forward neutral floor is to be kept: the central one lets more pileup in. Whether CMS's PUPPI already associates the charged candidates beyond 2.5 depends on its region configuration; in the genseed implementation it did not, and this one change is the largest improvement found in this study.

3.4 Dijets from every vertex

Two leading jets above 3 GeV (corrected), $\Delta\phi > 2.7$, a tag at $|\eta| < 1.305$, $\alpha = p_T^{(3)}/p_T^{\text{avg}} < 0.3$: 0.62 dijets per crossing from all vertices (`1v`), against 0.0047 at the primary vertex alone, where the forward junk takes the leading and the third jet (0.019 with `jva` or `none`). **Every pileup vertex is an independent minimum-bias dijet event**: 130 times the primary vertex's yield at low p_T . The L2Res closure, the mean $|R_{\text{MPF}}/r_{\text{true}} - 1|$ over the probe bins at $p_T^{\text{avg}} = 8\text{--}10$ GeV, is 0.03 without an α cut and 0.10 with $\alpha < 0.3$ (statistics) for `1v`, `none` and `ojet`; beyond $|\eta| = 2.5$ no probe bin reaches 50 dijets, because PUPPI leaves no forward jets to probe. The JER from the

bisector: with both jets in the same $|\eta|$ bin (0.08 per crossing, 2100 in the sample), the dijet-balance width $\sqrt{(\text{RMS}_{\parallel}^2 - \text{RMS}_{\perp}^2)/2}$ matches the true resolution (rms/mean of $p_{\text{T}}^{\text{corr}}/p_{\text{T}}^{\text{linkz}}$) within 10% from $p_{\text{T}}^{\text{avg}} = 5$ GeV in the barrel; the missing- p_{T} version (MPFX) comes out $\sim 25\%$ low and never within 10%, presumably because the soft activity in $\vec{p}_{\text{T}}^{\text{miss}}$ is correlated with the dijet imbalance, not isotropic; the core-fit versions need more than 200 dijets per bin and are empty. A statistically useful low- p_{T} JER and L2Res map needs about ten times this sample.

4 Conclusions

1. A jet-vertex association of the forward, vertex-blind energy by global missing- p_{T} (or missing- H_{T}) balance does not work for 5–20 GeV forward jets at $\langle N_{\text{PU}} \rangle = 45$: the balance cannot see the 1.6 GeV per vertex that belongs to single interactions. It may still help for hard forward jets in hard-scatter events, which this sample cannot test.
2. The forward spike at $2.5 < |\eta| < 3.0$ is mostly pileup made of several interactions. What removes it is information on the particles, not on the jets: PUPPI's vertex association extended to the charged candidates up to $|\eta| = 3$ reduces it by a factor 25–80. This should be tried in CMSSW.
3. Dijets from every vertex are the way to low- p_{T} JER and L2Res: two orders of magnitude more balanced dijets than the primary vertex, and a bisector JER that agrees with the truth from 5 GeV. The missing- p_{T} projections (MPF, MPFX) need care at these momenta. L3Res requires Z+jet events with pileup.

Code. `jvassoc.h` (optimiser; `jva.h` cannot coexist with `JVA.h` on a case-insensitive disk), `JVA.h/.C` (analyzer), `runJVA.C`, `runjva.sh` (knobs; any knob via `JVAOPT="key=value,..."`), `drawJVA.C` (this appendix), `scanJVA.C` (offline re-optimisation from the tuple), `cmpPuppiFwd.C` (the PUPPI comparison).

A Generated tables and figures (drawJVA.C, default PUPPI)

Table 1: Missing transverse momentum per method and vertex class: vertices (components / 2), in-axis rms and core σ of the components, mean $|\vec{p}_T^{\text{miss}}|$, and rms and core σ of the residual against the owner interaction's true imbalance (owned vertices); in GeV. *fail*: the core fails the protocol.

method	vertices	N	rms	σ_{core}	$\langle \vec{p}_T^{\text{miss}} \rangle$	rms _{res}	$\sigma_{\text{core, res}}$
dup	all vertices	856527	14.90	12.94	17.85	14.96	13.00
	PV	25000	17.03	15.90	20.83	16.79	15.53
	other vertices	831527	14.83	12.82	17.76	14.90	12.94
lv	all vertices	856527	7.18	4.43	7.75	7.48	4.89
	PV	25000	17.03	15.90	20.83	16.79	15.53
	other vertices	831527	6.66	4.39	7.35	6.97	4.74
trk	all vertices	856527	7.49	4.64	8.09	7.77	4.99
	PV	25000	16.65	15.36	20.34	16.45	15.23
	other vertices	831527	7.03	4.60	7.72	7.31	4.93
jva	all vertices	856527	6.89	4.44	7.57	7.20	4.89
	PV	25000	13.22	12.33	16.20	13.16	12.34
	other vertices	831527	6.61	4.39	7.31	6.92	4.74
ojet	all vertices	856527	6.92	4.44	7.60	7.20	4.89
	PV	25000	12.96	12.16	15.91	12.81	12.07
	other vertices	831527	6.66	4.40	7.35	6.93	4.74
opart	all vertices	856527	6.88	4.45	7.58	7.13	4.89
	PV	25000	12.76	11.96	15.67	12.58	11.78
	other vertices	831527	6.63	4.40	7.33	6.88	4.73
none	all vertices	856527	6.96	4.44	7.62	7.27	4.89
	PV	25000	13.55	12.39	16.50	13.48	12.40
	other vertices	831527	6.66	4.39	7.35	6.97	4.74

Table 2: The forward spike: jets per owned vertex at $2.5 < |\eta| < 3.0$ above 5, 10 and 20 GeV (reconstructed: p_T^{raw}) and their ratio to the pure generated jets of the owner interaction; the mean own-interaction fraction of the linked p_T there, and the fraction of those jets whose dominant interaction has less than 0.4, and more than 0.6, of the linked p_T ($p_T^{\text{raw}} > 5$ GeV).

method	jets per owned vertex, $p_T >$			reco/gen, $p_T >$			own	dominant share	
	5	10	20	5	10	20		< 0.4	> 0.6
dup	3.7757	1.1178	0.1316	22.48	49.61	64.72	0.034	0.732	0.061
lv	0.1153	0.0345	0.0039	0.69	1.53	1.93	0.129	0.700	0.074
trk	0.1329	0.0407	0.0046	0.79	1.80	2.26	0.168	0.695	0.073
jva	0.0042	0.0023	0.0004	0.03	0.10	0.21	0.132	0.700	0.097
ojet	0.0173	0.0054	0.0012	0.10	0.24	0.58	0.613	0.009	0.534
opart	0.0182	0.0033	0.0005	0.11	0.15	0.24	0.786	0.009	0.850
none	0.0000	0.0000	0.0000	0.00	0.00	0.00	0.347	0.250	0.250
pure gen	0.1680	0.0225	0.0020	1	1	1	1	–	–

Table 3: Fate of the forward clusters above 5 GeV under the methods that assign whole clusters, with and without track candidates: the number of clusters and the fraction of each outcome; *correct* is right plus combination nulled. The last rows are the reference of dropping every cluster (**none**, or derived from the combinations where the file has no **none**), whose correct fraction is the fraction of combinations.

method	tracks	N	right	wrong	single nulled	combo nulled	combo kept	correct
lv	yes	65178	0.037	0.106	0.000	0.000	0.857	0.037
	no	60520	0.081	0.259	0.000	0.000	0.660	0.081
trk	yes	65178	0.031	0.112	0.000	0.000	0.857	0.031
	no	60520	0.081	0.259	0.000	0.000	0.660	0.081
jva	yes	65178	0.003	0.005	0.134	0.816	0.042	0.818
	no	60520	0.003	0.003	0.334	0.655	0.004	0.659
ojet	yes	65178	0.143	0.000	0.000	0.857	0.000	1.000
	no	60520	0.340	0.000	0.000	0.660	0.000	1.000
none	yes	65178	0.000	0.000	0.143	0.857	0.000	0.857
	no	60520	0.000	0.000	0.340	0.660	0.000	0.660

Table 4: Dijet selection per crossing and method, for all vertices and for the primary vertex alone: the counts after each step divided by the number of crossings, cumulative up to $\alpha < 0.3$ (which requires a tag); *samebin* counts the pairs that pass $\Delta\phi$ and $\alpha < 0.3$ with both jets in the same $|\eta|$ bin, tag or not (the sample of the JER), so it can exceed the $\alpha < 0.3$ column.

method	vertices	vertex events	≥ 2 jets	$\Delta\phi$	tag	$\alpha < 0.3$	samebin
dup	all vertices	34.26	34.15	5.573	0.829	0.01268	0.03728
	PV	1	0.9992	0.2328	0.1477	0.00472	0.0006
lv	all vertices	34.26	21.6	3.634	2.645	0.6246	0.0828
	PV	1	0.9992	0.2328	0.1477	0.00472	0.0006
trk	all vertices	34.26	22.1	3.709	2.616	0.6082	0.0772
	PV	1	0.999	0.2429	0.1737	0.00892	0.00116
jva	all vertices	34.26	21.61	3.73	2.743	0.6466	0.08448
	PV	1	0.9976	0.2588	0.2096	0.01896	0.00232
ojet	all vertices	34.26	21.65	3.72	2.718	0.638	0.0832
	PV	1	0.9981	0.2678	0.2098	0.01796	0.00212
opart	all vertices	34.26	21.68	3.722	2.715	0.6348	0.08272
	PV	1	0.9981	0.2673	0.2108	0.01808	0.00216
none	all vertices	34.26	21.59	3.653	2.703	0.6377	0.08452
	PV	1	0.9976	0.2516	0.2055	0.01776	0.00232

Table 5: L2Res closure: the mean of $|R/r_{\text{true}} - 1|$ over the probe $|\eta|$ bins in which both are measured (their number in parentheses), per α selection, $p_{\text{T}}^{\text{avg}}$ bin, estimator and method, for the dijets of all vertices and of the primary vertex alone; only rows with a number are listed. A mean or median needs 50 effective dijets.

vertices	α	$p_{\text{T}}^{\text{avg}}$ [GeV]	R	dup	lv	trk	jva	ojet	opart	none
all vertices	< 0.3	5–6	R_{DB}	–	0.089 (11)	0.144 (13)	0.090 (11)	0.095 (11)	0.091 (11)	0.090 (11)
			R_{MPF}	–	0.059 (11)	0.104 (13)	0.059 (11)	0.062 (11)	0.060 (11)	0.059 (11)
		8–10	R_{DB}	–	0.127 (11)	0.264 (12)	0.124 (11)	0.125 (11)	0.134 (11)	0.124 (11)
			R_{MPF}	–	0.106 (11)	0.239 (12)	0.103 (11)	0.104 (11)	0.115 (11)	0.103 (11)
		12–15	R_{DB}	–	0.223 (2)	0.317 (1)	0.231 (2)	0.243 (2)	0.295 (1)	0.231 (2)
			R_{MPF}	–	0.250 (2)	0.290 (1)	0.257 (2)	0.263 (2)	0.284 (1)	0.257 (2)
	none	5–6	R_{DB}	0.257 (2)	0.061 (11)	0.102 (13)	0.061 (11)	0.061 (11)	0.058 (11)	0.061 (11)
			R_{MPF}	0.034 (2)	0.038 (11)	0.084 (13)	0.039 (11)	0.040 (11)	0.041 (11)	0.038 (11)
		8–10	R_{DB}	0.103 (15)	0.030 (11)	0.084 (13)	0.070 (13)	0.040 (13)	0.050 (13)	0.031 (11)
			R_{MPF}	0.074 (15)	0.033 (11)	0.087 (13)	0.069 (12)	0.030 (13)	0.076 (13)	0.032 (11)
		12–15	R_{DB}	0.097 (17)	0.045 (12)	0.057 (13)	0.074 (13)	0.056 (13)	0.050 (13)	0.044 (11)
			R_{MPF}	0.095 (17)	0.063 (12)	0.060 (13)	0.120 (13)	0.055 (13)	0.069 (12)	0.056 (11)
		18–21	R_{DB}	0.079 (14)	0.068 (12)	0.071 (13)	0.075 (11)	0.087 (12)	0.069 (11)	0.071 (10)
			R_{MPF}	0.080 (14)	0.098 (12)	0.087 (13)	0.113 (11)	0.081 (12)	0.083 (11)	0.084 (10)
		28–32	R_{DB}	0.365 (1)	–	0.083 (1)	0.097 (1)	0.113 (1)	0.113 (1)	0.063 (1)
			R_{MPF}	0.203 (1)	–	0.081 (1)	0.100 (1)	0.101 (1)	0.104 (1)	0.068 (1)
PV	none	8–10	R_{DB}	–	–	–	0.088 (3)	0.119 (2)	0.033 (1)	0.088 (3)
			R_{MPF}	–	–	–	0.062 (3)	0.086 (2)	0.041 (1)	0.062 (3)
		12–15	R_{DB}	0.068 (5)	0.068 (5)	0.057 (10)	0.079 (10)	0.075 (10)	0.072 (10)	0.079 (10)
			R_{MPF}	0.068 (4)	0.068 (4)	0.052 (10)	0.097 (10)	0.091 (10)	0.073 (10)	0.096 (10)
		18–21	R_{DB}	0.071 (7)	0.071 (7)	0.085 (9)	0.080 (10)	0.078 (10)	0.074 (10)	0.083 (10)
			R_{MPF}	0.059 (6)	0.059 (6)	0.067 (9)	0.068 (10)	0.063 (10)	0.058 (10)	0.069 (10)

Table 6: Reach of the dijet JER with the core-fit widths (against the truth $\sigma_{\text{core}}/\mu_{\text{core}}$): the lowest $p_{\text{T}}^{\text{avg}}$ (lower bin edge, GeV) with $|\text{JER}/\text{JER}_{\text{true}} - 1| < 0.1$ whose next two measured bins are not significantly outside that band, per $|\eta|$ bin and, below the line, per region of whole bins, per method and estimator (DB bisector, MPFX), for the same-bin dijets of all vertices and, in the last two columns, of the primary vertex under lv (the standard reconstruction). *none*: measured, never within the band; –: no $p_{\text{T}}^{\text{avg}}$ bin with both the JER and the truth. Only the rows with a measurement are listed.

$ \eta $	dup		lv		trk		jva		ojet		opart		none		PV, lv	
	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX
$0 < \eta < 1.305$	–	–	–	–	–	none	–	–	–	none	–	none	–	–	–	–
$1.305 < \eta < 2.5$	–	–	–	none	–	none	–	none	–	none	–	none	–	none	–	–

Table 7: Reach of the dijet JER with the plain-rms widths (against the truth rms/mean): the lowest $p_{\text{T}}^{\text{avg}}$ (lower bin edge, GeV) with $|\text{JER}/\text{JER}_{\text{true}} - 1| < 0.1$ whose next two measured bins are not significantly outside that band, per $|\eta|$ bin and, below the line, per region of whole bins, per method and estimator (DB bisector, MPFX), for the same-bin dijets of all vertices and, in the last two columns, of the primary vertex under lv (the standard reconstruction). *none*: measured, never within the band; –: no $p_{\text{T}}^{\text{avg}}$ bin with both the JER and the truth. Only the rows with a measurement are listed.

$ \eta $	dup		lv		trk		jva		ojet		opart		none		PV, lv	
	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX	DB	MPFX
$0 < \eta < 1.305$	–	–	5	none	5	none	5	none	5	none	5	none	5	none	–	–
$1.305 < \eta < 2.5$	–	–	none	none	none	none	none	none	none	none	none	none	none	none	–	–

Table 8: Summary per method, vertex level: rms and core σ of the missing- p_T components over all vertices, the core σ of their residual against the owner interaction’s true imbalance (GeV); the reconstructed over generated jets per owned vertex at $2.5 < |\eta| < 3.0$ above 10 GeV, and the mean own-interaction fraction of the linked p_T of the jets there above 5 GeV; the correct fraction of the forward clusters above 5 GeV (right vertex plus combination nulled), against which the last row is the reference of dropping every cluster.

method	rms	σ_{core}	$\sigma_{\text{core,res}}$	reco/gen $_{>10}$	own	correct
dup	14.90	12.94	13.00	49.61	0.034	–
lv	7.18	4.43	4.89	1.53	0.129	0.058
trk	7.49	4.64	4.99	1.80	0.168	0.055
jva	6.89	4.44	4.89	0.10	0.132	0.741
ojet	6.92	4.44	4.89	0.24	0.613	1.000
opart	6.88	4.45	4.89	0.15	0.786	–
none	6.96	4.44	4.89	0.00	0.347	0.762

Table 9: Summary per method, dijet level: same-bin dijets per crossing (all vertices / primary vertex); the L2Res closure, the mean $|R_{\text{MPF}}/r_{\text{true}} - 1|$ over the probe bins at p_T^{avg} 8–10 GeV (all vertices, with $\alpha < 0.3$ and without the α cut; number of probe bins in parentheses); and the reach of the core-fit MPFX JER (GeV) in the four regions of whole $|\eta|$ bins, all vertices / primary vertex under the method (*none*: measured, never within 10%; –: not measured).

method	dijets per crossing		closure		MPFX JER reach at $ \eta $, all / PV			
	all	PV	$\alpha < 0.3$	all α	0–1.305	1.305–2.5	2.5–3.139	3.139–5.191
dup	0.0373	0.0006	–	0.074 (15)	– / –	– / –	– / –	– / –
lv	0.0828	0.0006	0.106 (11)	0.033 (11)	– / –	none / –	– / –	– / –
trk	0.0772	0.0012	0.239 (12)	0.087 (13)	none / –	none / –	– / –	– / –
jva	0.0845	0.0023	0.103 (11)	0.069 (12)	– / –	none / –	– / –	– / –
ojet	0.0832	0.0021	0.104 (11)	0.030 (13)	none / –	none / –	– / –	– / –
opart	0.0827	0.0022	0.115 (11)	0.076 (13)	none / –	none / –	– / –	– / –
none	0.0845	0.0023	0.103 (11)	0.032 (11)	– / –	none / –	– / –	– / –

A.1 Missing transverse momentum

The missing transverse momentum of every usable vertex, $\vec{p}_T^{\text{miss}} = -\sum w_i \vec{p}_{T,i}$ over the candidates of its per-vertex event under each method, for all vertices, the primary vertex and the others (figure ??); its width against N_{PU} and against S_v , the scalar p_T sum of the vertex-resolved part (figure ??); and the residual against $-\sum \vec{p}_T$ of the generated particles of the owner interaction with $|\eta| < 5$, the imbalance a perfect detector would see. The rms is the in-axis rms; the core is a Gaussian iterated to convergence in $\mu \pm 2\sigma$, failed below 200 entries, without convergence, when it is wider than the distribution or off its median by more than $\sigma/2$. At the primary vertex `dup` and `lv` are the same event, the standard reconstruction. The sample has 25000 crossings at $\langle N_{\text{PU}} \rangle = 45$ with 34.3 usable vertices per crossing, 91% of them owned by an interaction (within 0.05 cm). Anti- k_T $R = 0.4$ from 1 GeV; candidates beyond $|\eta| = 2.5$ are vertex-blind; track candidates need 0.5 GeV in the cone; the optimiser uses $\sigma_v^2 = 1^2 + 1^2 S_v \text{ GeV}^2$ and the prices $\tau_{\text{trk}} = 4$, $\tau_{\text{all}} = 12$; the oracles accept a vertex within 0.2 cm. Jets are corrected with `text/jec_v1.txt`. PUPPI weights from `puppi.h` version `genseed-1.0`.

A.2 The forward spike

Jets of the per-vertex events of the owned vertices under each method, against the pure generated jets of the owner interaction (filled once per owned vertex): $dN/d|\eta|$ per owned vertex above three thresholds in p_T^{raw} (figure ??), the p_T^{corr} spectra in three regions (figure ??), and the purity of the jets above 5 GeV: the fraction of a jet's linked generated p_T that comes from the owner interaction, and the share of its dominant interaction (figure ??). A jet made of several interactions has a dominant share well below 1; the spike at $2.5 < |\eta| < 3.0$ is made of them.

A.3 Forward-cluster assignment

Every cluster of the vertex-blind candidates ($|\eta| > 2.5$, anti- k_T $R = 0.4$) is classified by the dominant interaction of its linked generated p_T : *single* when that share exceeds 0.5 and the interaction has a usable vertex within 0.2 cm, *combination* otherwise (unlinked clusters included). Its fate under each method that assigns whole clusters is then *right* (a single cluster sent to the vertex of its interaction), *wrong* (sent elsewhere), *single nulled*, *combination nulled* (the right thing to do with a combination) or *combination kept*; *correct* is right plus combination nulled. The clusters with track candidates (charged candidates of a vertex above 0.5 GeV in the cone at $|\eta| < 2.5$) and those without are shown apart: without tracks the optimiser can place a cluster at any vertex, at the price $\tau_{\text{all}} = 12$.

A.4 Relative response from every vertex

Dijets are selected from the per-vertex event of every usable vertex: the two leading jets in p_T^{corr} above 3 GeV at $|\eta| < 5.191$, $\Delta\phi > 2.7$, a tag at $|\eta| < 1.305$ and the other jet as the probe (both assignments at weight 1/2 when both are in the barrel), $\alpha = p_T^{(3)}/p_T^{\text{avg}}$. $R_{\text{DB}} = (1 + \langle A_{\text{DB}} \rangle)/(1 - \langle A_{\text{DB}} \rangle)$ with $A_{\text{DB}} = (p_T^{\text{probe}} - p_T^{\text{tag}})/(p_T^{\text{probe}} + p_T^{\text{tag}})$, and the same for $A_{\text{MPF}} = \vec{p}_T^{\text{miss},T1} \cdot \hat{p}_T^{\text{tag}}/(2p_T^{\text{avg}})$ (type-1 correction above 3 GeV). The closure reference r_{true} is the median of the per-event ratio of the probe's to the tag's $p_T^{\text{corr}}/p_T^{\text{linkz}}$, where p_T^{linkz} is the linked generated p_T of the jet. A mean or a median needs 50 effective dijets.

A.5 Jet energy resolution from dijets

Dijets whose two jets fall into the same $|\eta|$ bin, with $\alpha < 0.3$, are projected on the bisector axis $\hat{n} = (\hat{u}_1 - \hat{u}_2)/|\hat{u}_1 - \hat{u}_2|$, along which both jets make the same angle, and on its normal: $(\vec{p}_{T,1} + \vec{p}_{T,2}) \cdot \hat{n}/p_T^{\text{avg}}$ for the DB bisector and $\vec{p}_T^{\text{miss},T1} \cdot \hat{n}/p_T^{\text{avg}}$ for MPFX. The parallel width holds both jets' resolution and the imbalance along the axis, the perpendicular one the imbalance across it, so the resolution per jet is $\sigma = \sqrt{(\text{RMS}_{\parallel}^2 - \text{RMS}_{\perp}^2)/2}$; a bin with $\text{RMS}_{\perp} \geq \text{RMS}_{\parallel}$ has no JER and counts as a failed

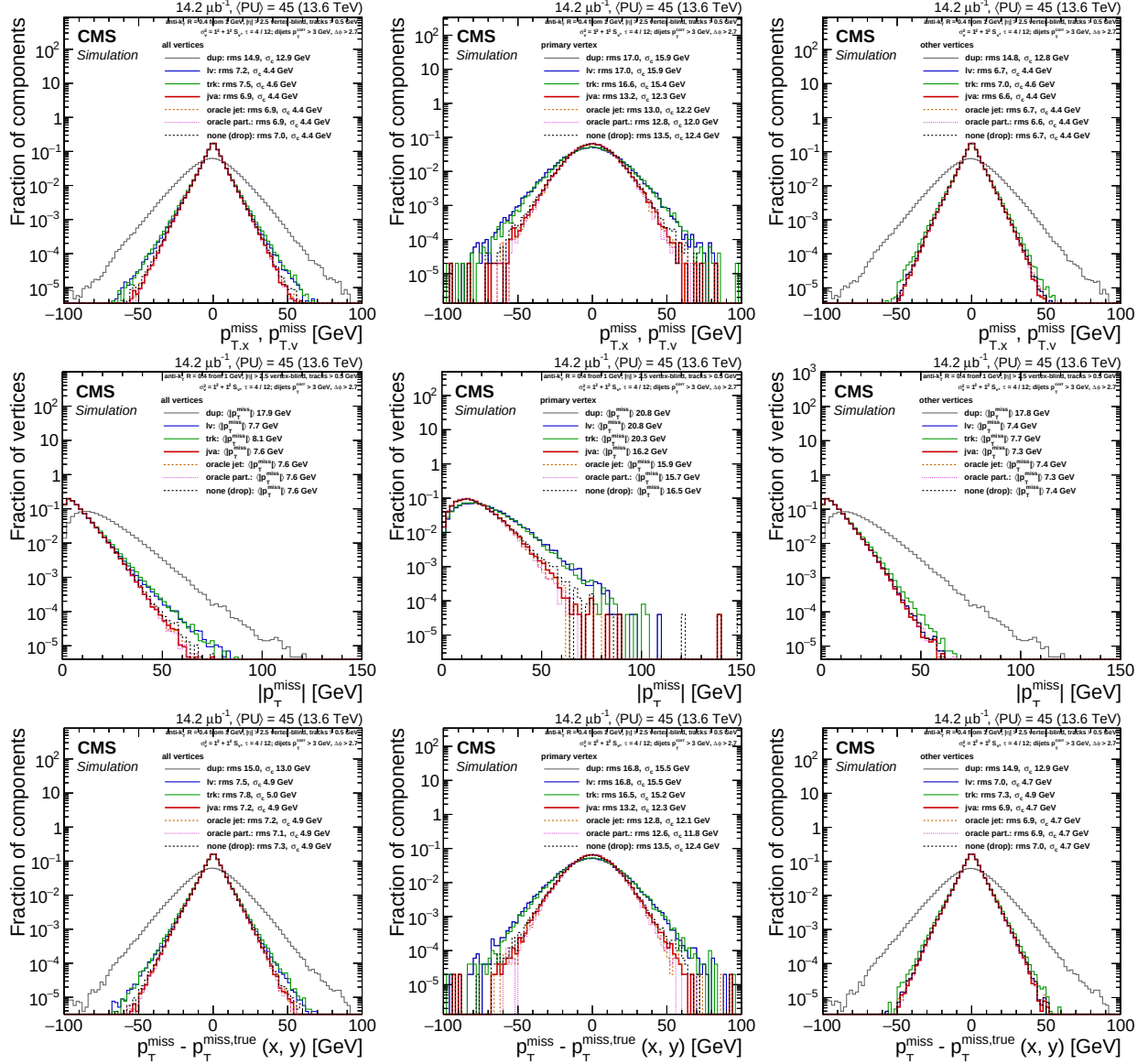


Figure 1: Missing transverse momentum per vertex, unit-normalised: first row the x and y components, second row the magnitude, third row the residual of the components against the owner interaction's true imbalance (owned vertices), for all vertices, the primary vertex and the other vertices (left to right), for **dup** (grey), **lv** (blue), **trk** (green), **jva** (red), **ojet** (open orange), **opart** (open magenta), **none** (open black and dashed). The legend quotes the rms and the core σ (or the mean magnitude). Over all vertices the rms of a component is **dup** 14.9 GeV, **lv** 7.2 GeV, **jva** 6.9 GeV, **opart** 6.9 GeV, **none** 7.0 GeV. The core width of the residual against the truth is **dup** 13.0 GeV, **lv** 4.9 GeV, **jva** 4.9 GeV, **opart** 4.9 GeV, **none** 4.9 GeV.

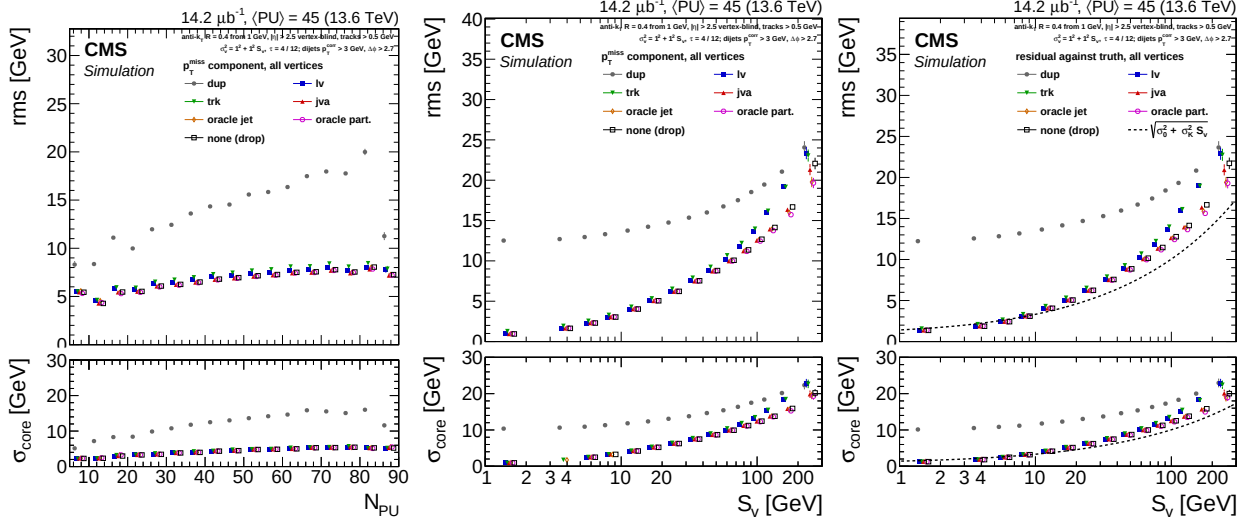


Figure 2: Width of the missing transverse momentum of all vertices: the components against N_{PU} and against S_v , and the residual against the owner interaction's true imbalance against S_v (owned vertices), with the noise model of the optimiser, $\sqrt{\sigma_0^2 + \sigma_K^2 S_v}$ (dashed). Upper panels the rms, lower panels the core σ ; methods as in figure ???. The model describes the vertex-resolved part against the owner's true central recoil; the residual here is against the true imbalance over $|\eta| < 5$ and so also holds the part of the owner's forward recoil that no forward cluster carries (PUPPI keeps little of it), which is why it lies above the model at large S_v . At S_v in 30–40 GeV the model gives 6.0 GeV, the residual core of `opart` is 7.4 GeV and of `jva` 7.4 GeV.

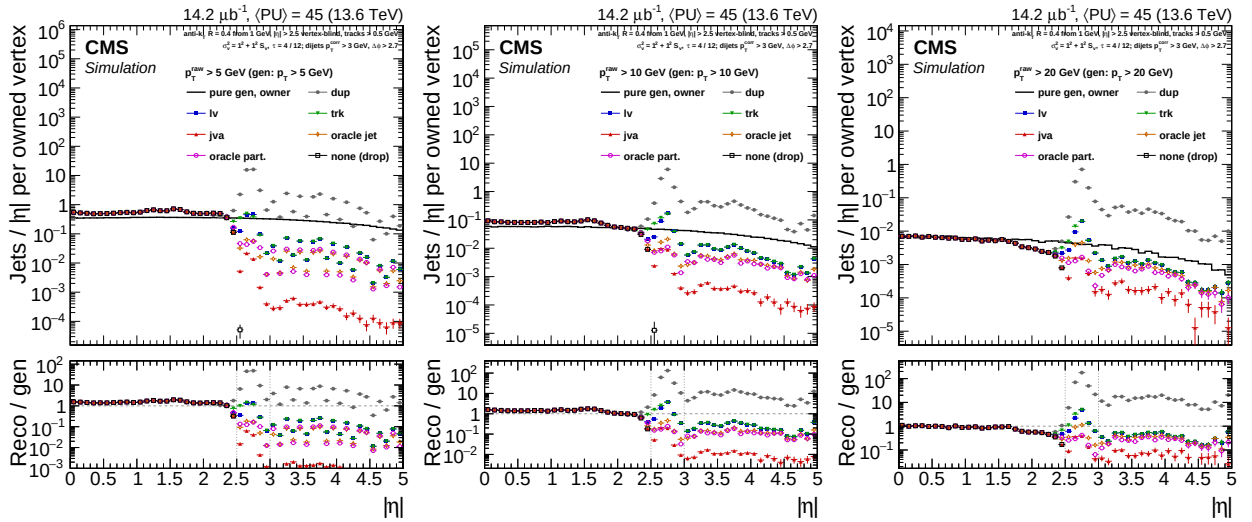


Figure 3: Jets per owned vertex against $|\eta|$ above 5, 10 and 20 GeV (left to right; reconstructed jets in p_T^{raw}): the pure generated jets of the owner interaction (black line) and the reconstructed jets of the owned vertex's event under `dup` (grey), `lv` (blue), `trk` (green), `jva` (red), `ojet` (open orange), `opart` (open magenta), `none` (open black and dashed), with the ratio to the generated jets below (on a logarithmic scale where the methods span more than a factor 20); the dotted lines mark $2.5 < |\eta| < 3.0$. At $2.5 < |\eta| < 3.0$ above 10 GeV the reconstructed jets per owned vertex are 49.61 (`dup`), 1.53 (`lv`), 1.80 (`trk`), 0.10 (`jva`), 0.24 (`ojet`), 0.15 (`opart`), 0.00 (`none`) times the generated ones.

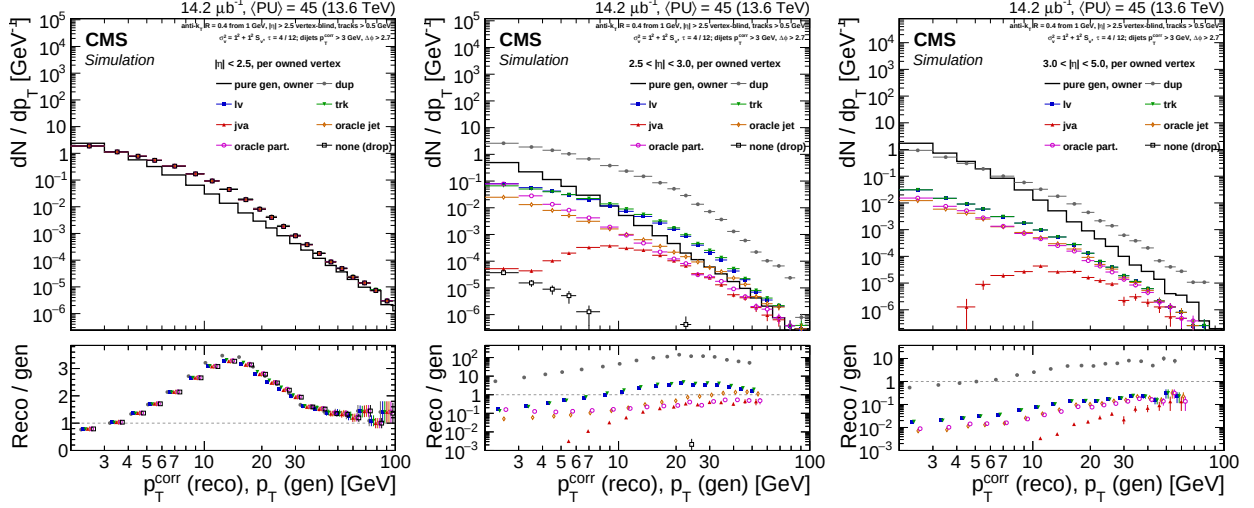


Figure 4: p_T spectra per owned vertex of the pure generated jets of the owner interaction and of the reconstructed jets (p_T^{corr} , corrected to the genseed link scale) under each method, at $|\eta| < 2.5$, $2.5 < |\eta| < 3.0$ and $3.0 < |\eta| < 5.0$ (left to right), with the ratio to the generated spectrum below (on a logarithmic scale where the methods span more than a factor 20). At $2.5 < |\eta| < 3.0$ and 10–12 GeV reco/gen is 45.92 (dup), 1.43 (lv), 1.75 (trk), 0.06 (jva), 0.20 (ojet), 0.18 (opart), not defined (none).

bin. Each distribution is first mirror-symmetrised ($h(x) + h(-x)$: the sign of $\hat{n}$ follows the order of the two jets, and a p_T -ordered pair folds the parallel projection at zero); the width is then the core σ of a Gaussian centred at zero, iterated in $\pm 2\sigma$ (primary), or the plain in-axis rms. Each needs 200 dijets per p_T^{avg} bin, on a p_T^{avg} axis of 2 merged fine bins from 3 GeV (3–5, 5–8, 8–12, 12–18 GeV, ...). The truth is the width of $p_T^{\text{corr}}/p_T^{\text{linkz}}$ of both jets of the same dijets, like with like: $\sigma_{\text{core}}/\mu_{\text{core}}$ under the response protocol for the core-fit JER, rms/mean for the plain-rms JER. The *reach* is the lowest p_T^{avg} bin with $|\text{JER}/\text{JER}_{\text{true}} - 1| < 0.1$ whose next two measured bins are not significantly outside that band ($|\text{JER}/\text{JER}_{\text{true}} - 1| < 0.1 + 2\sigma$, and no failed bin). Besides the 18 $|\eta|$ bins, the same analysis is made in four regions of whole bins (barrel, endcap, transition, HF), whose same-bin samples are the sums of their bins' (figure ?? and figure ??).

A.6 Summary

The key number of each section per method is collected in table ?? (vertex level) and table ?? (dijets).

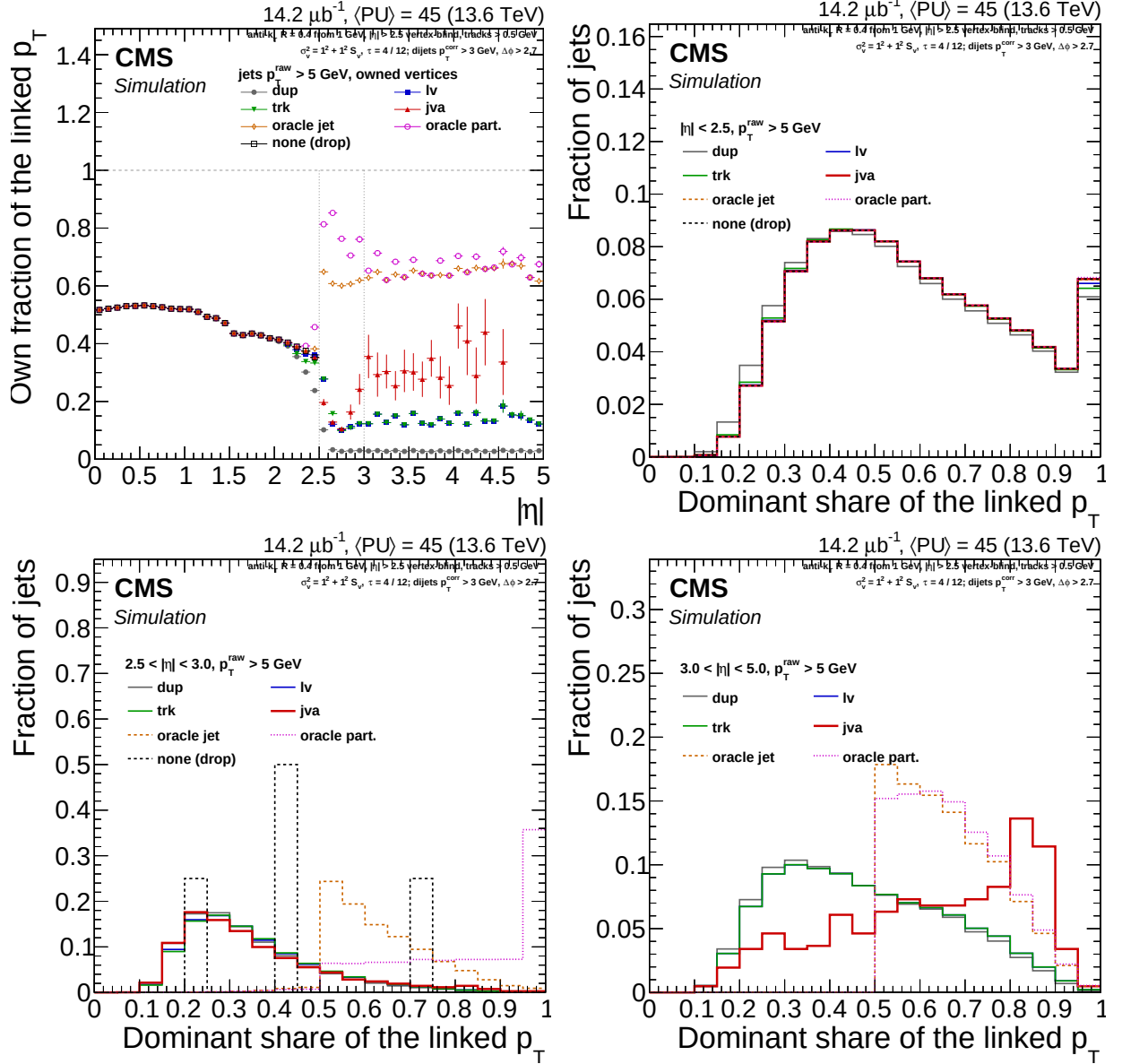


Figure 5: Purity of the jets above 5 GeV at the owned vertices: the fraction of the linked generated p_T that comes from the owner interaction against $|\eta|$ (the profile), and the distribution of the dominant interaction's share of the linked p_T at $|\eta| < 2.5$, $2.5 < |\eta| < 3.0$ and $3.0 < |\eta| < 5.0$ (a share of exactly 1 is in the last bin). The mean own fraction at $2.5 < |\eta| < 3.0$ is dup 0.03, lv 0.13, trk 0.17, jva 0.13, ojet 0.61, opart 0.79, none 0.35. Fraction of the jets at $2.5 < |\eta| < 3.0$ with a dominant share below 0.4 / above 0.6: dup 0.73 / 0.06, lv 0.70 / 0.07, trk 0.70 / 0.07, jva 0.70 / 0.10, ojet 0.01 / 0.53, opart 0.01 / 0.85, none 0.25 / 0.25.

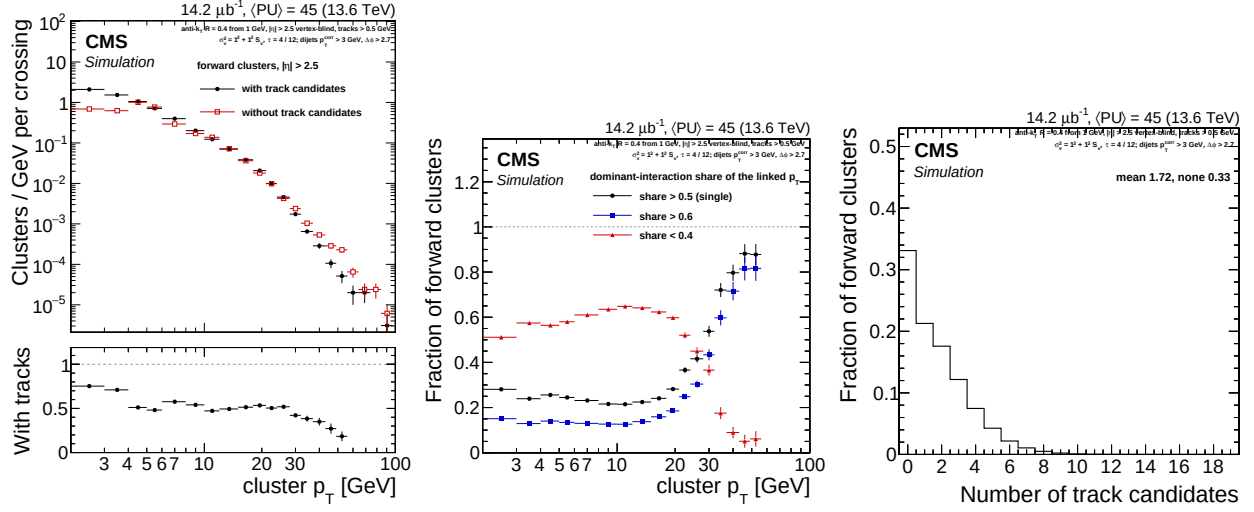


Figure 6: The forward clusters: their p_T spectrum per crossing with and without track candidates and the fraction with tracks, the fraction whose dominant interaction has more than 0.5, more than 0.6 and less than 0.4 of the linked p_T (an unlinked cluster has share 0), and the number of track candidates. Above 5 GeV there are 5.03 clusters per crossing, 52% of them with track candidates. At 10–12 GeV 0.21 of the clusters are single (share > 0.5) and 0.65 have a share below 0.4. A cluster has 1.72 track candidates on average; 33% have none.

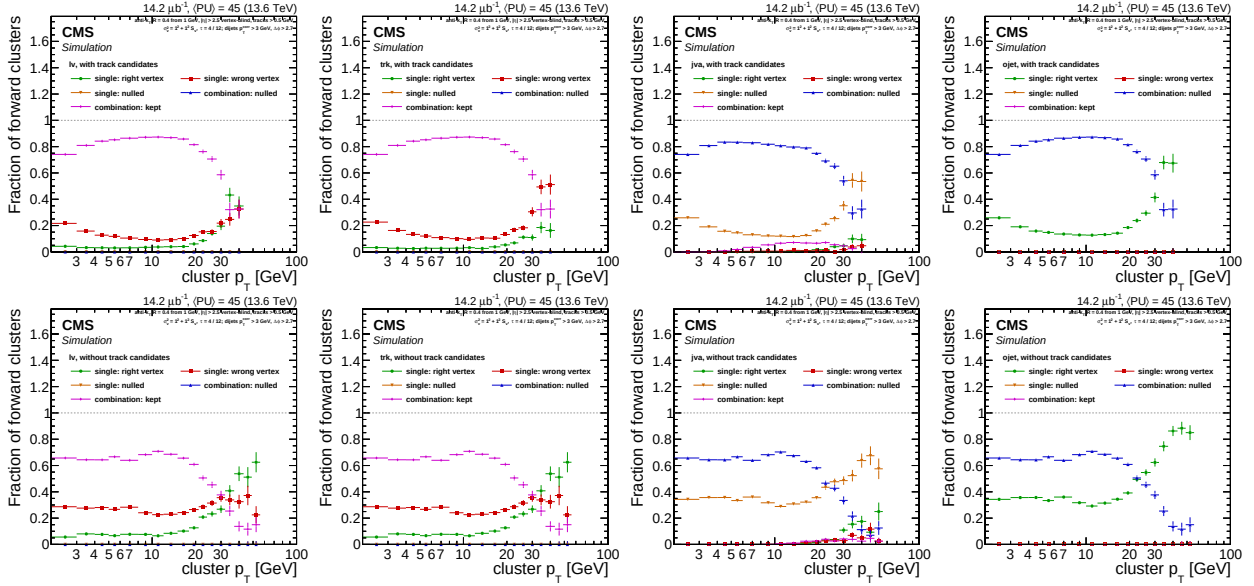


Figure 7: Outcome fractions of the forward clusters against the cluster p_T for lv, trk, jva and ojet (left to right), for the clusters with track candidates (first row) and without (second row): single cluster at the right vertex (green), at a wrong vertex (red), nulled (orange); combination nulled (blue) or kept (magenta). Above 5 GeV the correct fraction is lv 0.06, trk 0.05, jva 0.74, ojet 1.00, none 0.76 (clusters with and without tracks together), where none, dropping every cluster, scores the fraction of combinations. The outcomes add up to the cluster spectrum in every bin (largest relative difference 0).

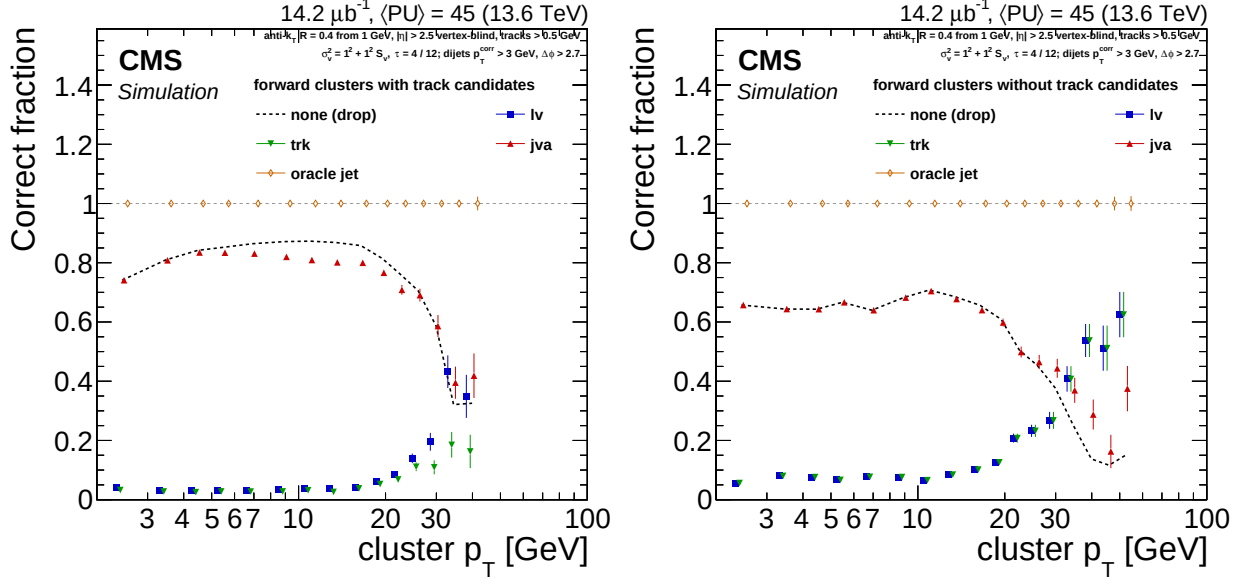


Figure 8: The correct fraction (right vertex plus combination nulled) of the forward clusters against the cluster p_T , with track candidates (left) and without (right), per method, and the reference of dropping every cluster (**none**, black dashed), which scores the fraction of combinations: a method that places clusters helps only where it lies above that line. Above 5 GeV with tracks: **lv** 0.037, **trk** 0.031, **jva** 0.818, **ojet** 1.000; **none** 0.857. Above 5 GeV without tracks: **lv** 0.081, **trk** 0.081, **jva** 0.659, **ojet** 1.000; **none** 0.660.

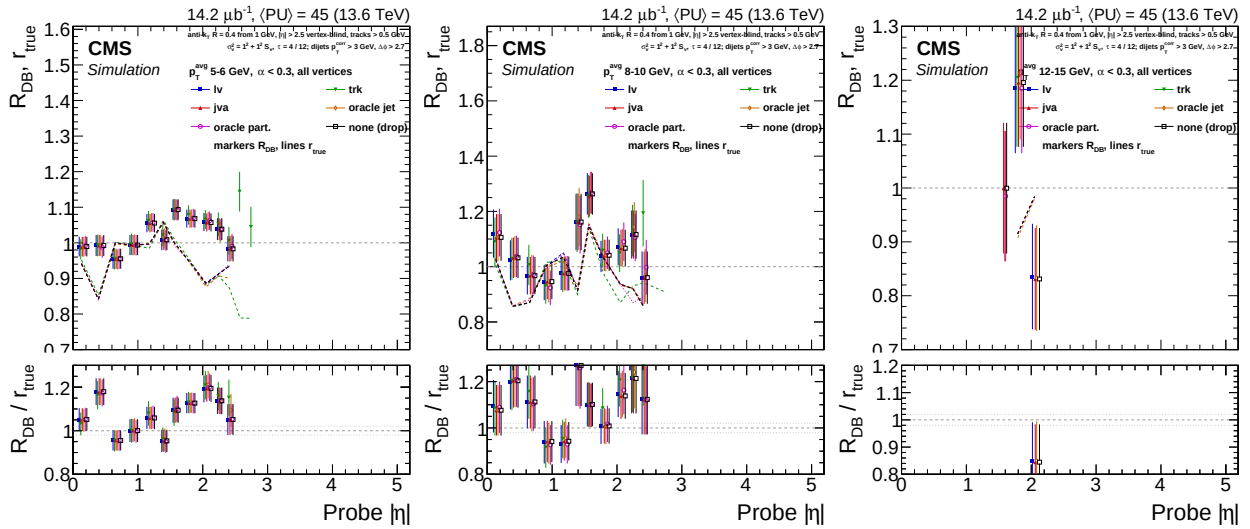


Figure 9: R_{DB} against the probe $|\eta|$ for the dijets of all vertices with $\alpha < 0.3$, at p_T^{avg} 5–6, 8–10 and 12–15 GeV (left to right, top to bottom; of the five L2Res bins, those in which some method has two probe bins with 50 effective dijets), per method (markers, slightly shifted in $|\eta|$ for legibility), with the truth relative response r_{true} of the same dijets (lines, same colour); below, the closure R/r_{true} , with $\pm 2\%$ dotted. The mean $|R_{DB}/r_{\text{true}} - 1|$ over the probe bins where both are measured (their number in parentheses) is, at 5–6 GeV **lv** 0.089 (11), **trk** 0.144 (13), **jva** 0.090 (11), **ojet** 0.095 (11), **opart** 0.091 (11), **none** 0.090 (11); at 8–10 GeV **lv** 0.127 (11), **trk** 0.264 (12), **jva** 0.124 (11), **ojet** 0.125 (11), **opart** 0.134 (11), **none** 0.124 (11); at 12–15 GeV **lv** 0.223 (2), **trk** 0.317 (1), **jva** 0.231 (2), **ojet** 0.243 (2), **opart** 0.295 (1), **none** 0.231 (2).

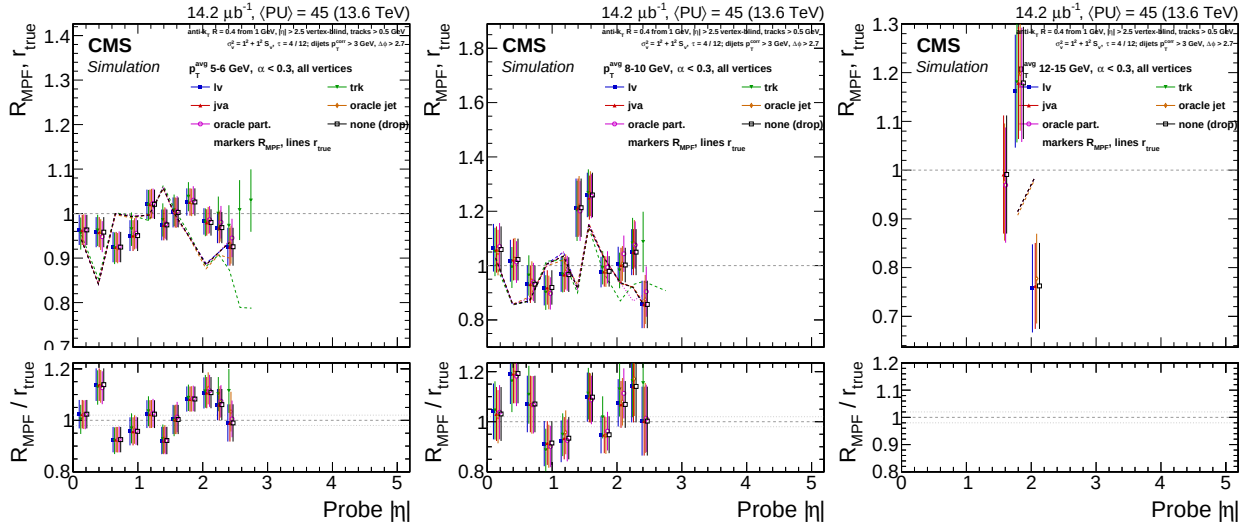


Figure 10: R_{MPF} against the probe $|\eta|$ for the dijets of all vertices with $\alpha < 0.3$, at $p_{\text{T}}^{\text{avg}}$ 5–6, 8–10 and 12–15 GeV (left to right, top to bottom; of the five L2Res bins, those in which some method has two probe bins with 50 effective dijets), per method (markers, slightly shifted in $|\eta|$ for legibility), with the truth relative response r_{true} of the same dijets (lines, same colour); below, the closure R/r_{true} , with $\pm 2\%$ dotted. The mean $|R_{\text{MPF}}/r_{\text{true}} - 1|$ over the probe bins where both are measured (their number in parentheses) is, at 5–6 GeV lv 0.059 (11), trk 0.104 (13), jva 0.059 (11), ojet 0.062 (11), opart 0.060 (11), none 0.059 (11); at 8–10 GeV lv 0.106 (11), trk 0.239 (12), jva 0.103 (11), ojet 0.104 (11), opart 0.115 (11), none 0.103 (11); at 12–15 GeV lv 0.250 (2), trk 0.290 (1), jva 0.257 (2), ojet 0.263 (2), opart 0.284 (1), none 0.257 (2).

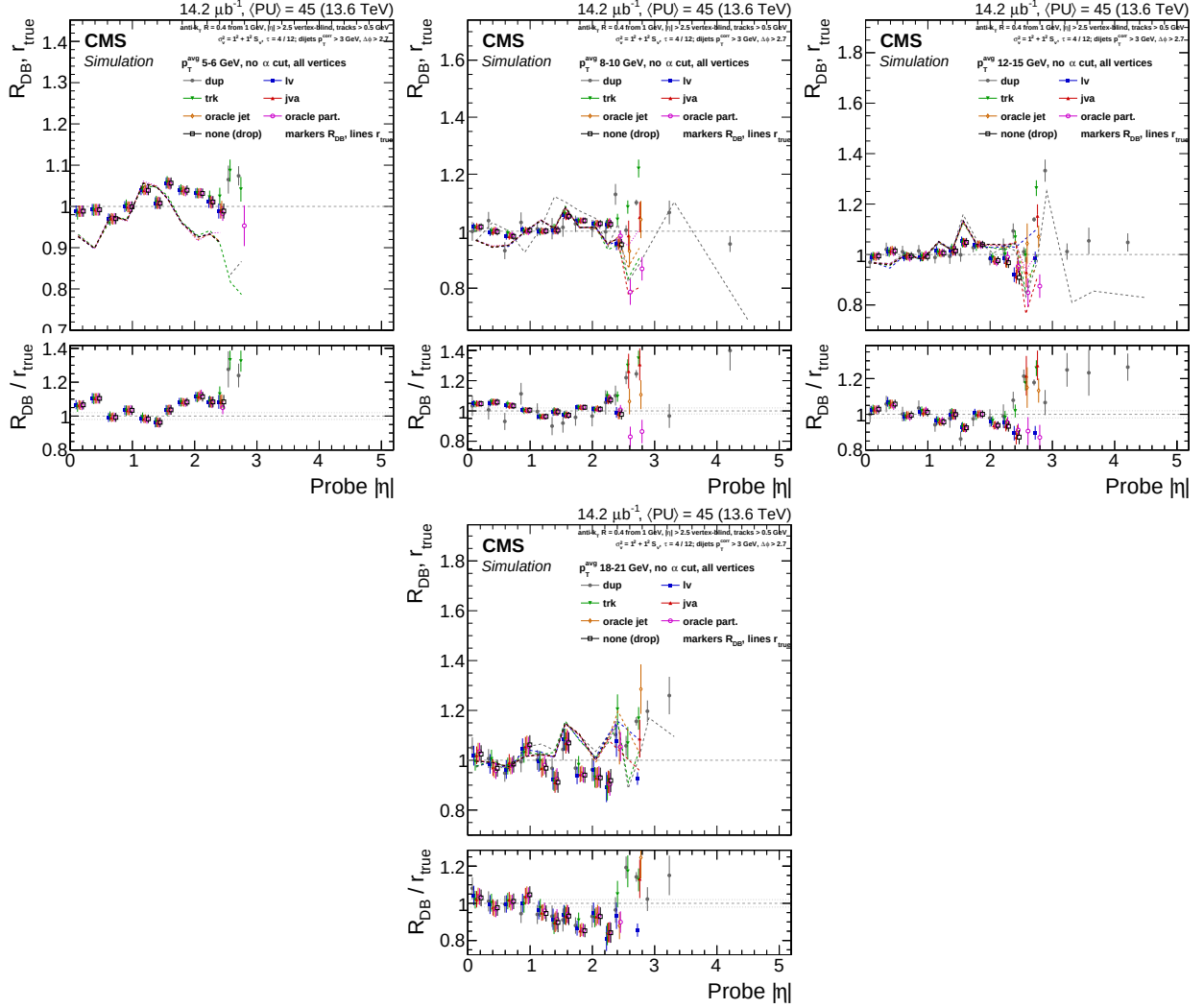


Figure 11: R_{DB} against the probe $|\eta|$ for the dijets of all vertices without the α cut (the $\alpha < 0.3$ selection is in figure ?? and figure ??), at $p_{\text{T}}^{\text{avg}}$ 5–6, 8–10, 12–15 and 18–21 GeV (left to right, top to bottom; of the five L2Res bins, those in which some method has two probe bins with 50 effective dijets), per method (markers, slightly shifted in $|\eta|$ for legibility), with the truth relative response r_{true} of the same dijets (lines, same colour); below, the closure R/r_{true} , with $\pm 2\%$ dotted. The mean $|R_{\text{DB}}/r_{\text{true}} - 1|$ over the probe bins where both are measured (their number in parentheses) is, at 5–6 GeV dup 0.257 (2), lv 0.061 (11), trk 0.102 (13), jva 0.061 (11), ojet 0.061 (11), opart 0.058 (11), none 0.061 (11); at 8–10 GeV dup 0.103 (15), lv 0.030 (11), trk 0.084 (13), jva 0.070 (13), ojet 0.040 (13), opart 0.050 (13), none 0.031 (11); at 12–15 GeV dup 0.097 (17), lv 0.045 (12), trk 0.057 (13), jva 0.074 (13), ojet 0.056 (13), opart 0.050 (13), none 0.044 (11); at 18–21 GeV dup 0.079 (14), lv 0.068 (12), trk 0.071 (13), jva 0.075 (11), ojet 0.087 (12), opart 0.069 (11), none 0.071 (10).

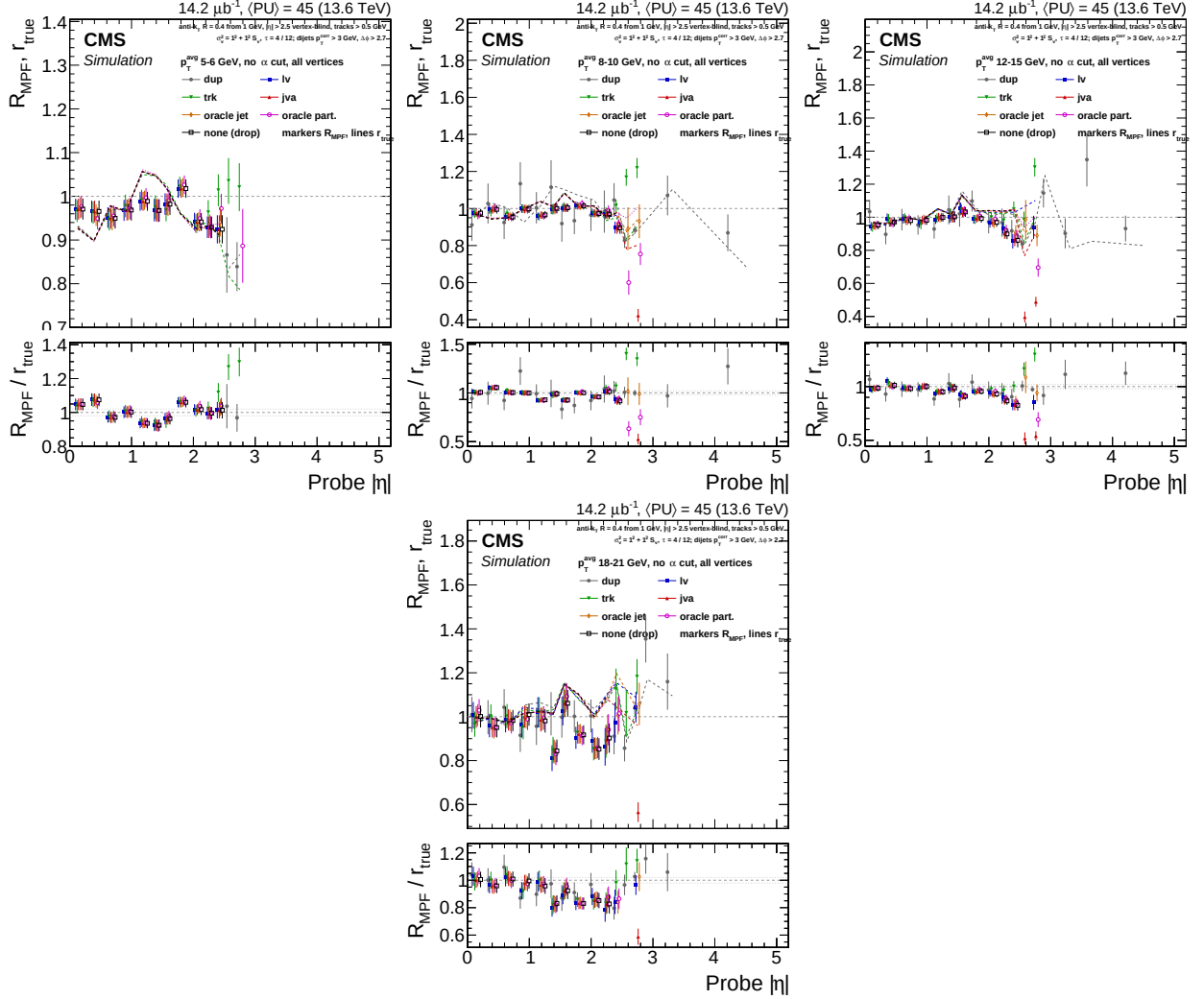


Figure 12: R_{MPF} against the probe $|\eta|$ for the dijets of all vertices without the α cut (the $\alpha < 0.3$ selection is in figure ?? and figure ??), at p_T^{avg} 5–6, 8–10, 12–15 and 18–21 GeV (left to right, top to bottom); of the five L2Res bins, those in which some method has two probe bins with 50 effective dijets), per method (markers, slightly shifted in $|\eta|$ for legibility), with the truth relative response r_{true} of the same dijets (lines, same colour); below, the closure R/r_{true} , with $\pm 2\%$ dotted. The mean $|R_{\text{MPF}}/r_{\text{true}} - 1|$ over the probe bins where both are measured (their number in parentheses) is, at 5–6 GeV dup 0.034 (2), lv 0.038 (11), trk 0.084 (13), jva 0.039 (11), ojet 0.040 (11), opart 0.041 (11), none 0.038 (11); at 8–10 GeV dup 0.074 (15), lv 0.033 (11), trk 0.087 (13), jva 0.069 (12), ojet 0.030 (13), opart 0.076 (13), none 0.032 (11); at 12–15 GeV dup 0.095 (17), lv 0.063 (12), trk 0.060 (13), jva 0.120 (13), ojet 0.055 (13), opart 0.069 (12), none 0.056 (11); at 18–21 GeV dup 0.080 (14), lv 0.098 (12), trk 0.087 (13), jva 0.113 (11), ojet 0.081 (12), opart 0.083 (11), none 0.084 (10).

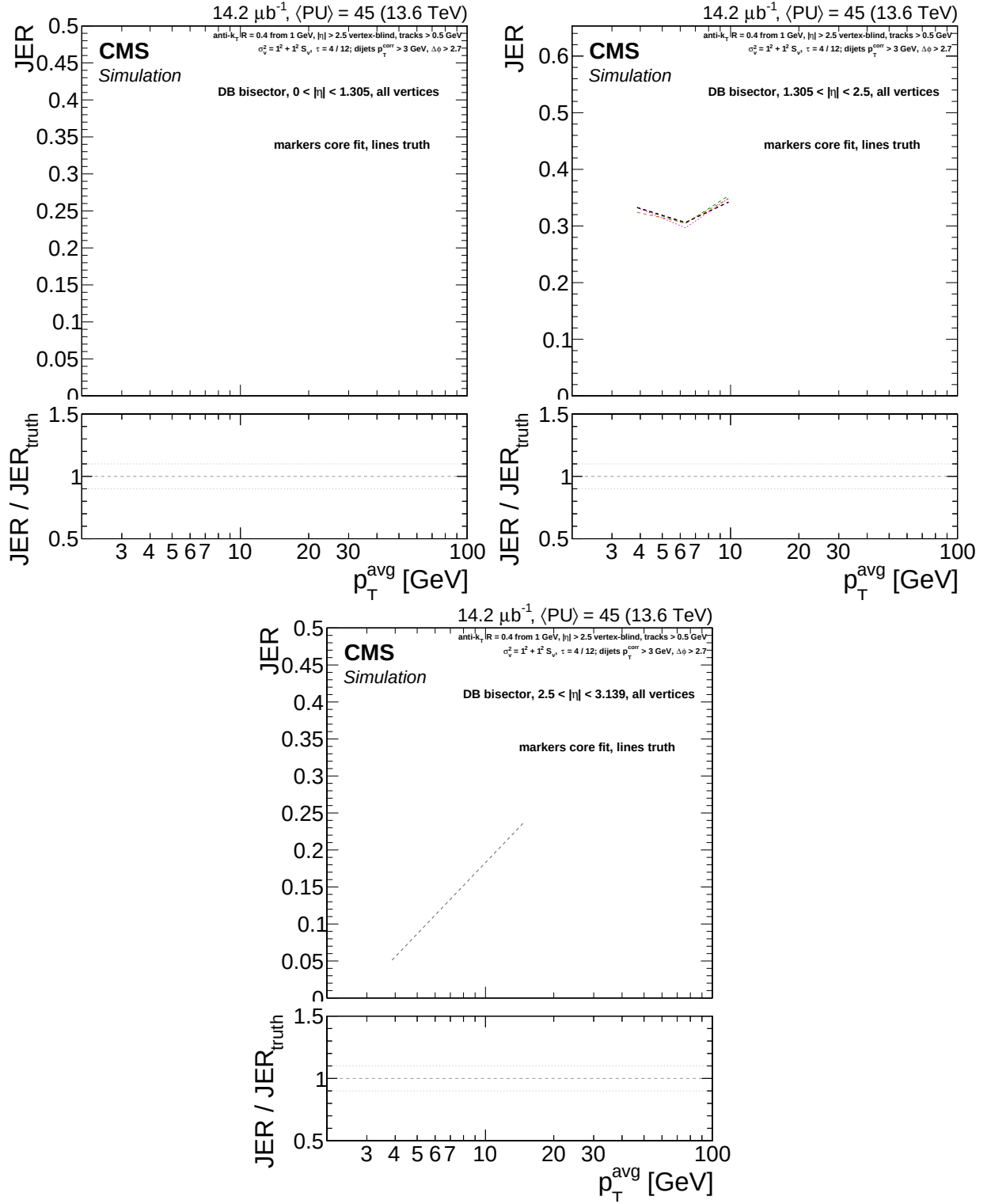


Figure 13: JER per jet from the DB bisector method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$) in four regions made of whole $|\eta|$ bins, $0 < |\eta| < 1.305$, $1.305 < |\eta| < 2.5$, $2.5 < |\eta| < 3.139$ and $3.139 < |\eta| < 5.191$ (left to right, top to bottom; the regions without 200 dijets in any p_T^{avg} bin are left out): core-fit JER per method (markers) and the truth $\sigma_{\text{core}}/\mu_{\text{core}}$ of $p_T^{\text{corr}}/p_T^{\text{linkz}}$ of the same jets (lines, same colour); below, the ratio with $\pm 10\%$ dotted, and a cross on the bottom edge where $\text{RMS}_{\perp} \geq \text{RMS}_{\parallel}$.

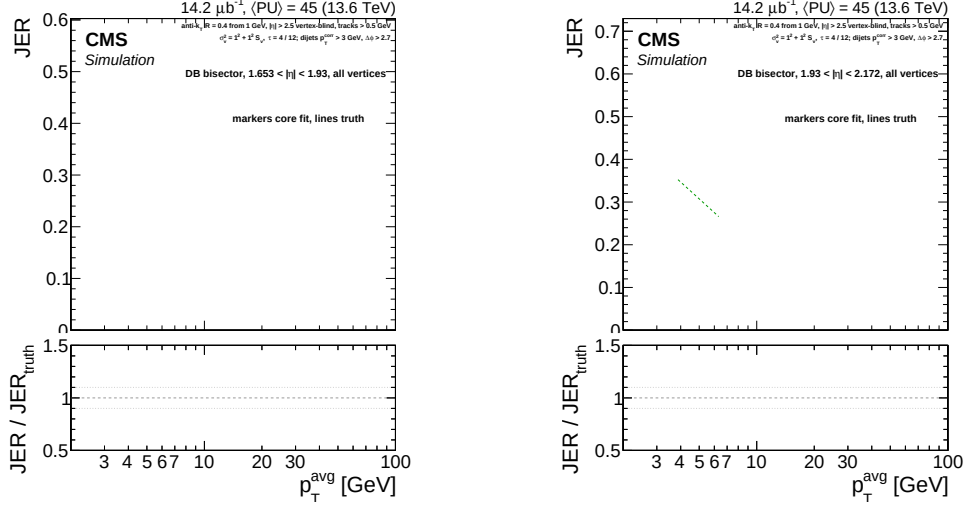


Figure 14: JER per jet from the DB bisector method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$), per $|\eta|$ bin from $0 < |\eta| < 0.261$ to $1.93 < |\eta| < 2.172$ (left to right, top to bottom; the bins without 200 dijets in any p_T^{avg} bin are left out); markers, lines and the lower panels as in figure ??.

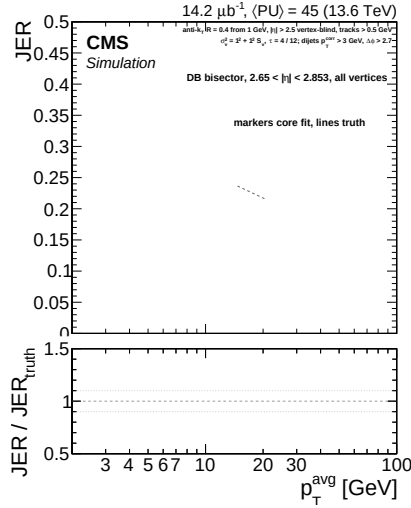


Figure 15: JER per jet from the DB bisector method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$), per $|\eta|$ bin from $2.172 < |\eta| < 2.322$ to $3.839 < |\eta| < 5.191$ (left to right, top to bottom; the bins without 200 dijets in any p_T^{avg} bin are left out); markers, lines and the lower panels as in figure ??.

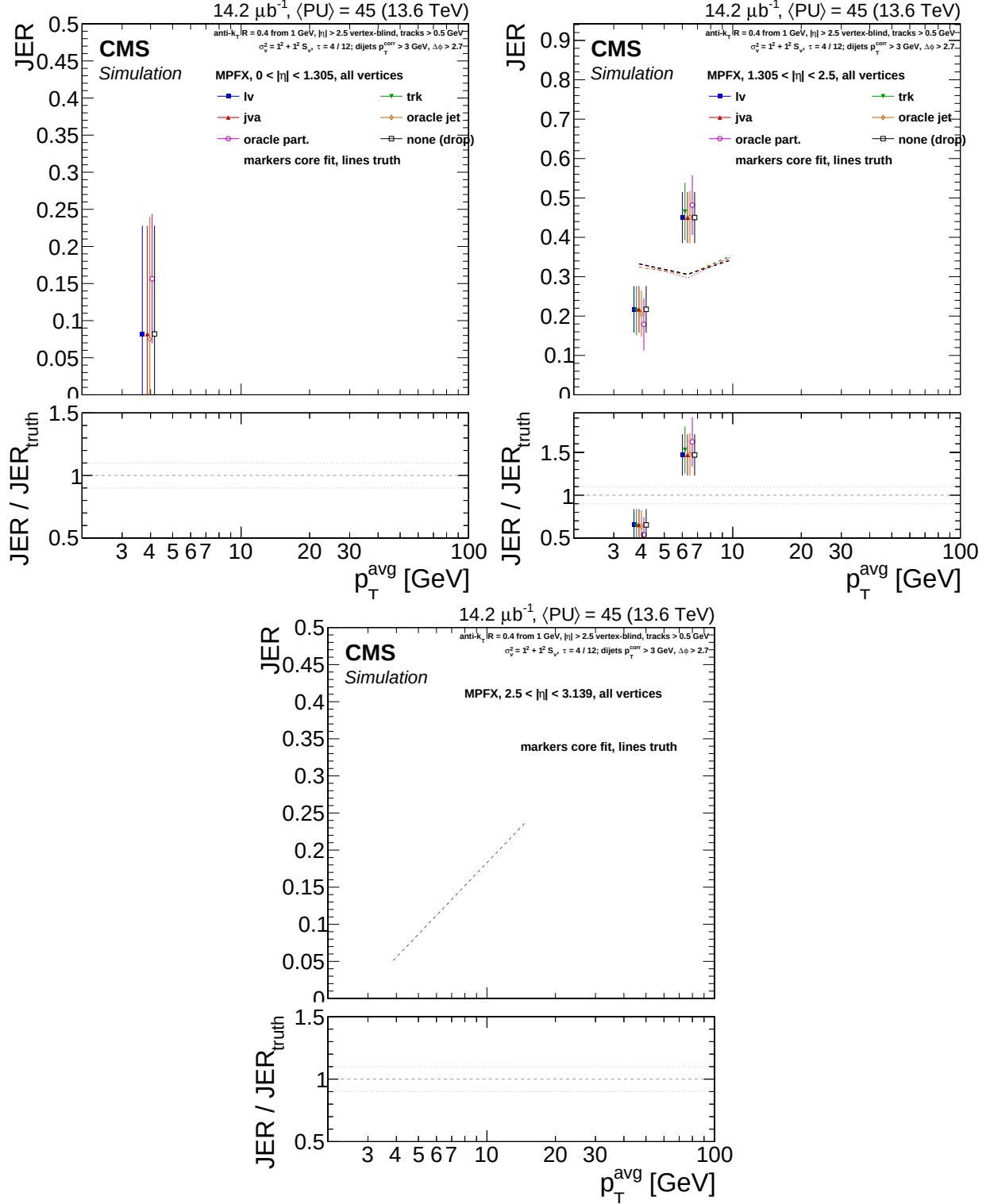


Figure 16: JER per jet from the MPFX method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$) in four regions made of whole $|\eta|$ bins, $0 < |\eta| < 1.305$, $1.305 < |\eta| < 2.5$, $2.5 < |\eta| < 3.139$ and $3.139 < |\eta| < 5.191$ (left to right, top to bottom; the regions without 200 dijets in any p_T^{avg} bin are left out): core-fit JER per method (markers) and the truth $\sigma_{\text{core}}/\mu_{\text{core}}$ of $p_T^{\text{corr}}/p_T^{\text{linkz}}$ of the same jets (lines, same colour); below, the ratio with $\pm 10\%$ dotted, and a cross on the bottom edge where $\text{RMS}_{\perp} \geq \text{RMS}_{\parallel}$. Reach at $0 < |\eta| < 1.305$: **opart** none. Reach at $1.305 < |\eta| < 2.5$: **lv** none, **jva** none, **opart** none, **none** none.

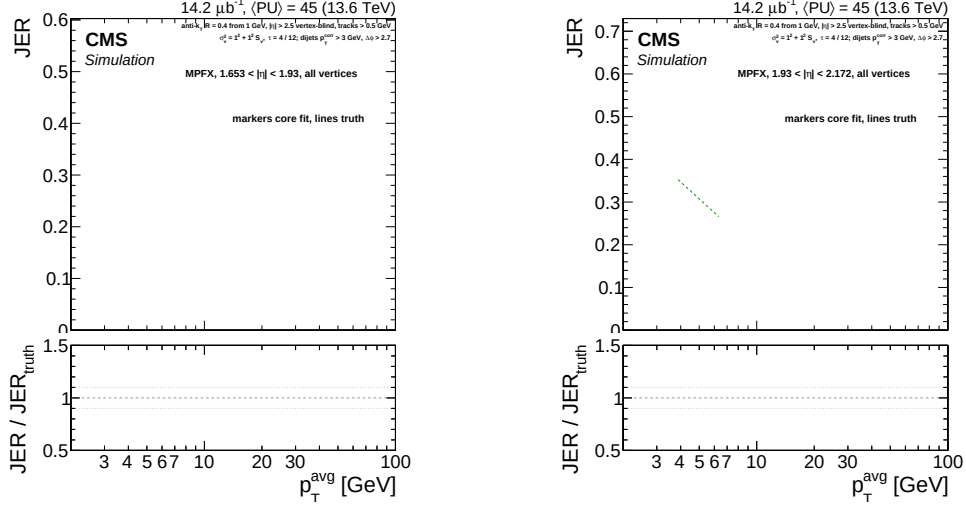


Figure 17: JER per jet from the MPFX method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$), per $|\eta|$ bin from $0 < |\eta| < 0.261$ to $1.93 < |\eta| < 2.172$ (left to right, top to bottom; the bins without 200 dijets in any p_T^{avg} bin are left out); markers, lines and the lower panels as in figure ??.

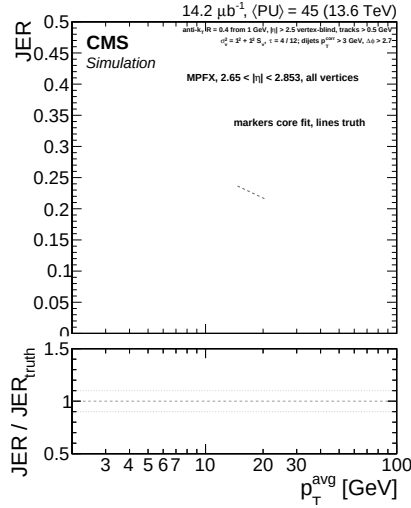


Figure 18: JER per jet from the MPFX method against p_T^{avg} for the same-bin dijets of all vertices ($\alpha < 0.3$), per $|\eta|$ bin from $2.172 < |\eta| < 2.322$ to $3.839 < |\eta| < 5.191$ (left to right, top to bottom; the bins without 200 dijets in any p_T^{avg} bin are left out); markers, lines and the lower panels as in figure ??.